

Homework 5

“React Games with Animation”

The following problem set is worth 100 points total. Please submit a zip file containing all relevant code to the Canvas drop box by the due date of the assignment.

Requirements

For the remainder of the semester, we will be discussing a very popular JavaScript framework called React. This assignment involves building an application using the framework.

“React Games” (100 pts)

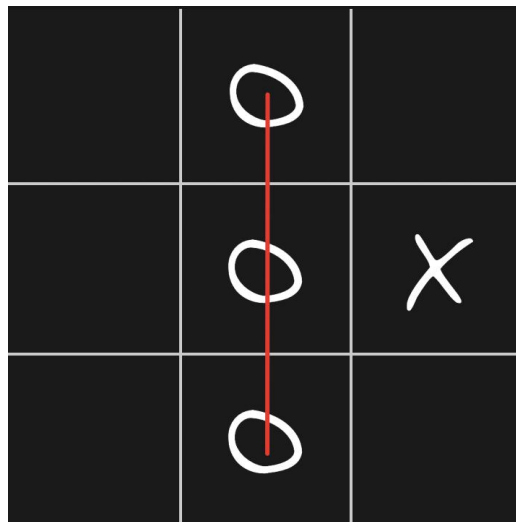
For this assignment, you will be implementing a pair of classic games using React.

1. Tic Tac Toe
2. Snake Game

The only stipulation is that you will add a couple of twists to your games.

Tic-Tac-Toe

The idea of the game is the following:



Tic-Tac-Toe

Objective:

Be the first player to get **three of your marks in a row** — horizontally, vertically, or diagonally — on a 3x3 grid.

Players:

- Two players.
 - One plays **X**, the other plays **O**.
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Gameplay:

1. The game starts with an **empty 3x3 grid**.
 2. Players take **turns** placing their mark (X or O) in an **empty square**.
 3. The **first player** is usually X, but this can be decided however you like.
 4. Players alternate turns until:
 - One player gets **3 in a row** — **wins**, or
 - All squares are filled and no one wins — it's a **draw**.
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Winning:

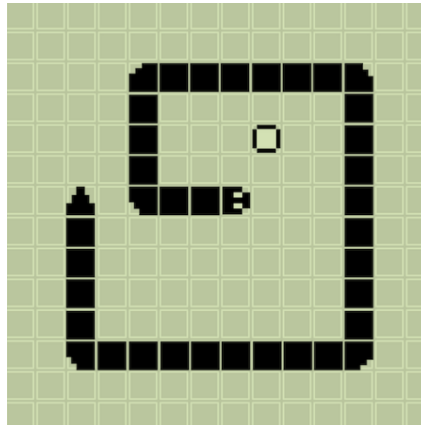
You win if you get **three of your symbols** in:

- A **row** (like top row: X | X | X),
 - A **column** (like middle column: O | O | O), or
 - A **diagonal** (from top-left to bottom-right or top-right to bottom-left).
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Draw:

If all 9 squares are filled and **no one has won**, the game is a **tie** or **draw**.

Snake Game



Snake Game

Rules of the snake game are as follows:

Objective

- Control a snake to **eat food** and **grow longer**.
- Avoid **colliding** with yourself or the walls (depending on the version).

Basic Controls

- Use arrow keys (or WASD) to move the snake:
➡ Right, ⬅ Left, ⬆ Up, ⬇ Down
- The snake **cannot reverse direction directly** (e.g., if you're going right, you can't go left immediately).

Gameplay Rules

1. **Start Small:** The snake starts at a fixed length (usually 1–3 segments).
2. **Eat Food:** A piece of food appears randomly on the grid. When the snake eats it:
 - The snake grows by 1 segment.
 - Your score increases.
 - A new piece of food appears.
3. **No Collisions:**
 - Hitting the wall = Game Over (in most versions).
 - Hitting your own tail = Game Over.

4. **Continuous Movement:** The snake moves continuously in the current direction until you change it.
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Winning or Losing

- You **lose** when you crash into a wall or into yourself.
- There's usually **no winning condition**, but some versions let you win if you fill the screen.

Detailed Requirements

These two games are standard JavaScript games you can easily find on the internet. The difference with this assignment is that you will implement both using React. You can use either JavaScript or TypeScript. Some more detailed descriptions:

- For Tic-Tac-Toe, your opponent is the computer, and the computer randomly selects *the best square* during their turn
- In the Tic-Tac-Toe game either the user or the computer can go first
- For the snake game, you will need to consider how often food spawns and how to use the keyboard to control the snake in React

Animation Libraries

The other half of this assignment is to incorporate an animation library into your application. The most popular library today is called **React Spring** and there are numerous examples you can use:

<https://react-spring.dev/examples>

Of course you can use another library of your choice, but for this assignment you have many options of things you can animate. For instance, in Tic-Tac-Toe when a mark is placed on the board or the general movement of the snake itself.

React Router

The last piece of this assignment is to link your two games together using a navigation bar with **React Router**. This is similar to the routing done in both Vue and Angular.

Submission Requirements

For submission, upload a zip file containing all assets I need to run your React application. Do not include the **node_modules** directory in your submission. Feel free to use either JavaScript or TypeScript in your implementation. Submit using the Canvas drop box before the due date.